

MotionBlind: Probing the Illusion of Motion Understanding in Video-LLMs

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Abstract

Video large language models (Video-LLMs) are increasingly used as the perceptual front end of world models, a role that assumes they can read *motion*: how fast something moves, which way it travels, how hard it is pushed. We show they cannot. A Video-LLM can watch two clips of the same person in the same room, name every object in both, and still fail to say which clip moves faster. We introduce **MotionBlind**, a contrastive benchmark of self-recorded video for physically grounded motion (speed, magnitude, and direction), the variables a world model must predict. Each instance is a pair of near-identical clips that differ only in motion. Each clip carries two complementary yes/no questions, giving four items per instance, and a model earns credit only if all four are correct. We report Instance Accuracy (I_{Acc}), which has a 6.25% chance floor. Single-frame, appearance, and language-only shortcuts all collapse to it. MotionBlind complements the recent TimeBlind benchmark. We run a controlled study of six open and two frontier Video-LLMs, varying whether the video is present, whether frames are shown in the correct temporal order, and how frames are sampled (1 to 24 frames, four selection strategies). Open models sit near the 6.25% floor, and scale does not help. Removing the video drops every model to zero I_{Acc} , and shuffling frames collapses I_{Acc} to chance, so the task genuinely needs video in order. Neither more frames nor smarter frame selection closes the gap, because these change *which* frames are seen, not whether motion is read. Only Gemini 3.1 Pro clears the benchmark overall, and even it fails on speed. A front end that cannot tell two speeds of the same action apart is not yet a trustworthy source of supervision, reward, or evaluation for a world model.



MotionBlind Dataset



ostadabbas/MotionBlind



Project Page

1 Introduction

Show a video large language model (Video-LLM) two short clips of the same person in the same room, differing only in that one walks left and the other walks right, and it will describe both scenes fluently (the person, the clothing, the furniture) and then, as often as not, get the direction wrong. The static content is read perfectly; the motion is not. This is the pattern that our MotionBlind benchmark is built to expose (Figure 1), while striking at a role Video-LLMs are increasingly asked to play.

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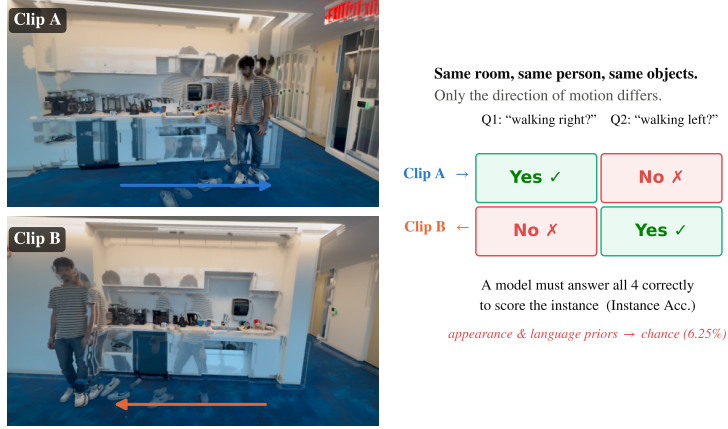


Figure 1: The illusion of motion understanding. A contrastive MotionBlind instance: two self-recorded clips share the same room, person, and objects and differ *only* in the direction of motion (frames overlaid stop-motion style, faint→solid, to visualize the trajectory). Each clip carries two complementary yes/no questions, and a model scores the instance only if all four items are correct (I_{Acc}). Any strategy that leans on static appearance or language priors (naming the objects, guessing the likely action) is driven to the 6.25% chance floor.

World models [10] predict how an environment evolves, and increasingly they lean on large multimodal models rather than physics engines alone. Video-LLMs now sit across this stack: auto-labeling state transitions for data curation, acting as reward or success detectors in model-based reinforcement learning, and grounding language instructions in an agent’s observations. Each role asks the model not to recognize *what* is in a scene but to perceive *how* it changes, direction, speed, duration, and the fine-grained kinematics a world model must ultimately predict.

This is a strong assumption, and a growing body of evidence suggests it does not hold. Benchmarks built around single video clips and free-form or multiple-choice questions consistently overestimate temporal competence, because a model can often answer correctly from a single representative frame, from object co-occurrence statistics, or from language priors about what typically happens next, without ever attending to motion [14, 15]. The fix that has emerged in recent work is a contrastive or minimal-pairs protocol, inspired by Winoground’s visio-linguistic compositionality test for images [25]: construct two clips that share identical static content and differ only in temporal structure, pair each with two complementary questions, and require a model to get all four combinations right. Vinoground [26] applied this idea to natural short videos and found that even GPT-4o barely exceeds chance on the strict group-level metric. Most directly relevant to us, the TimeBlind benchmark [13] scales this paradigm to 600 curated instances (2,400 video-question pairs) spanning three levels of temporal composition (atomic events, event properties, and event interdependencies) and shows that the best of over twenty frontier Multimodal Large Language Models (MLLMs) reaches only 48.2% Instance Accuracy, I_{Acc} ³. Related diagnostic work shows the same failure mode along other motion axes, including signed motion direction [12] and camera motion [6], reinforcing that this is a perceptual bottleneck rather than a benchmark artifact.

TimeBlind’s taxonomy, however, emphasizes semantic/compositional reasoning (state transitions, causal contingency, ordering). Two questions remain for physical AI: does the same collapse occur on physically-grounded properties (e.g., speed, magnitude, direction) isolated as controlled variables, and can anything be recovered purely by controlling which and how many frames a model sees, a lever every ingestion pipeline already has?

We introduce **MotionBlind**, a benchmark of 240 video-question pairs (60 contrastive 2×2 instances over 82 self-recorded clips) targeting physically grounded motion; speed, magnitude, and direction; that is hard to source and label from internet video but directly controllable in a lab (Figure 2). Where TimeBlind maps the breadth of temporal blindness, MotionBlind trades scope for controlled depth on the physical core of motion.

³ I_{Acc} is an instance-level accuracy metric: an instance is counted as correct only if all four predicted items are correct. This makes it a strict end-to-end evaluation metric with a chance level of 6.25%. Unless otherwise noted, I_{Acc} is our primary evaluation metric. For reference, humans achieve 98.2% on this metric on TimeBlind benchmark [13].



Figure 2: MotionBlind versus TimeBlind. (a) The self-recorded set: 60 contrastive instances (82 clips, 240 items) across three physical-motion characteristics—speed, magnitude, and direction—each with a representative, face-blurred frame. (b) Words most distinctive of each benchmark’s questions (weighted log-odds [16]): MotionBlind turns on manner, magnitude, and direction (*faster*, *farther*, *closer*, *clockwise*); TimeBlind on appearance and state change (*color*, *turns*, *top/bottom*, *gradually*). Manner terms appear in 94% of MotionBlind questions versus 36% of TimeBlind’s. (c) The same questions in the evaluated models’ backbone (Qwen2.5-VL-7B, mean-pooled), projected onto the axis between the two benchmarks’ means. The two are linearly separable (AUC 0.97); TimeBlind further splits into a self-recorded and a web-collected cluster, with MotionBlind adjacent to former.

Two design choices make the benchmark a clean diagnostic of motion understanding. First, all clips are self-recorded, with the objects, scene, and background held constant within each pair, ensuring that the only meaningful difference is the motion property under evaluation. Second, every relative-motion question compares two moving references within the same scene rather than relying on absolute concepts such as "fast," "slow," "hard," or "soft." This eliminates the need for context-dependent (operational) definitions, whose meanings vary with the scenario. For example, a speed considered slow for a moving car may be fast for a walking person.

We benchmark Video-LLMs on MotionBlind alongside two reference datasets: **TimeBlind** [13], which shares our contrastive 2×2 protocol but only targets semantic temporal reasoning rather than isolating physical motion (speed, direction, magnitude) as a controlled variable, and **Video-MME** [7], a standard non-contrastive video-QA suite that establishes the models we test are broadly capable and calibrates what our integrity probes do to a benchmark that does *not* require motion understanding. Our controlled diagnostic study evaluates Video-LLMs along three complementary axes. First, we assess video necessity using a no-video, text-only probe to determine whether the questions can be answered without visual input. Second, we evaluate temporal-order sensitivity using shuffled- and reversed-frame probes, which preserve the visual content of individual frames while disrupting their temporal order. Third, we study the effect of frame selection by comparing four strategies across frame budgets ranging from 1 to 24 frames: uniform sampling, random sampling, a GRPO-trained learned frame selector (HORNet [3], trained with GRPO [21]), and a training-free key-clip selector (Frame2Clip [22]). Together, these experiments test whether motion-centric questions can be solved without video, without correct temporal ordering, from a single frame, by increasing the number of sampled frames, or by employing more sophisticated frame-selection strategies.

We evaluate six open Video-LLMs spanning different design philosophies: Eagle2.5-8B [4], a long-context model; Qwen3-VL-4B [19], a recent general-purpose model; Motion-o (7B) [8], a motion-tuned variant of Qwen2.5-VL-7B [20]; Gemma-4-12B-it [23], a larger general-purpose model; Molmo2-8B [2, 5], an open video-native model; and Inkling-Small [24], a reasoning model. We further compare against two frontier proprietary models, Gemini 3.1 Pro [9] and GPT-5.6 Luna [18].

On MotionBlind, the task genuinely requires video in the correct order: text-only accuracy is exactly zero, and shuffling or reversing frames collapses I_{Acc} toward chance. Every open model stays near that floor regardless of scale, and neither more frames nor dynamic frame selection (learned or training-free) beats plain uniform sampling at matched budgets; adaptive selection changes *which* frames are seen, not whether model inherently understands the motion presented in a video sequence. Only the frontier Gemini 3.1 Pro clears MotionBlind overall, and even it fails on the purely rate-defined categories, isolating a representational gap that sampling strategies do not close.

We position MotionBlind as both a diagnostic benchmark and, ultimately, a training signal for the perceptual front end of Physical AI world models. A system that cannot reliably distinguish two visually identical actions performed at different speeds, magnitudes, or directions cannot serve as a trustworthy source of physically grounded supervision, reward, or evaluation. Our contributions are:

- MotionBlind, a contrastive, self-recorded benchmark for evaluating physically grounded motion perception through three fundamental motion attributes: speed, magnitude, and direction. MotionBlind complements TimeBlind’s semantically driven taxonomy while adopting the same strict I_{Acc} evaluation metric.
- A controlled diagnostic study that systematically isolates three factors in video understanding: (i) the necessity of video, (ii) sensitivity to temporal order, and (iii) frame-selection strategy. Across six open Video-LLMs, two frontier proprietary models, and frame budgets from 1 to 24, we show that the observed failures are fundamentally perceptual and order-dependent rather than purely linguistic, and persist even as model capability scales.
- An analysis of frame selection showing that neither a GRPO-trained learned selector nor a training-free key-clip selector consistently outperforms uniform sampling for pairwise motion discrimination. Our analysis reveals that current frame selection methods largely ignore temporal ordering and produce nearly uniform frame-importance scores, motivating future order-aware and discrimination-aware training objectives.

2 Related Work

Video-LLMs and their perceptual front ends. Modern Video-LLMs sample a fixed budget of frames, encode them with a vision tower, and condition a language model for captioning and QA. This "frame aggregation + LLM reasoning" paradigm drives gains on aggregate benchmarks [15], exemplified by long-context sampling in Eagle2.5-8B, the Qwen-VL line’s dynamic resolution [19, 20], and open video-native models such as Molmo/Molmo2 [2, 5]. We treat these as fixed front ends and probe what they perceive once static appearance is removed, rather than proposing an architecture. To isolate that residual, recent benchmarks use minimal pairs of near-identical videos (TempCompass [14], Vinoground [25, 26], TimeBlind [13]); we adopt TimeBlind’s I_{Acc} metric and 2×2 protocol and extend it toward physically grounded, controllable properties (speed, magnitude, direction).

Frame sampling and selection for Video-LLMs. Which frames a model sees strongly affects aggregate scores such as Video-MME [7]. Beyond uniform and random sampling, learned selectors score frames by informativeness for single-answer QA (e.g. HORNet [3], trained with GRPO), while training-free methods select temporally coherent *clips* to preserve motion (e.g. Frame2Clip [22]). We evaluate both and find their long-form gains do not transfer to pairwise, order-sensitive discrimination; a consequence of selectors whose reward and encoders carry no notion of temporal position or of the paired video.

World models predict how an environment evolves for planning and imagined rollouts [10, 11], and increasingly use pretrained multimodal models as data generators, labelers, or plausibility evaluators. The reliability of that component’s motion perception is therefore a precondition for the world model’s physical grounding, and MotionBlind is a targeted diagnostic for exactly this precondition [1, 17].

3 Introducing MotionBlind

MotionBlind is a self-recorded contrastive benchmark designed to isolate physically-grounded motion understanding from appearance, object identity, scene context, and language priors. Clips are recorded at 30 fps and run 5.9 s on average (median 4.5 s, range 1.9–15.7 s), at resolutions from 480×848 to 1280×1280, predominantly portrait. Unlike existing video benchmarks, every example is constructed so that the only meaningful difference between two clips is the motion being performed. Consequently, the correct answer cannot be inferred from a single frame, object recognition, or semantic context.

MotionBlind contains 60 contrastive instances (82 video clips and 240 question-answer items, with answers balanced 50/50 between yes and no). Each instance consists of a minimal pair of near-identical videos that share the same object, scene, background, camera viewpoint, and lighting while differing

only in one motion attribute. By controlling all other visual cues, MotionBlind ensures that motion is the only signal capable of determining the correct answer.

The benchmark focuses on three fundamental physical properties of motion: *speed* (7 instances), *magnitude* (27 instances), and *direction* (26 instances, including 18 translational and 8 rotational motions) (see Figure 2a). The paired clip serves only to define the comparison structure and is never required to determine the answer to a question.

Comparison Benchmarks. We evaluate MotionBlind alongside TimeBlind [13], which adopts the same contrastive 2×2 evaluation protocol but targets semantic temporal reasoning rather than physically grounded motion. TimeBlind contains 600 contrastive instances (2,400 question-answer items) collected from curated internet videos, simulations, and real-world recordings. While both datasets share an identical evaluation protocol, they probe complementary aspects of video understanding. MotionBlind asks *how* an action is performed by focusing on speed, magnitude, and direction, whereas TimeBlind asks *what* changes and *when*. This distinction is reflected in the language of the datasets, with motion-related terms appearing in 94% of MotionBlind questions compared with 36% of TimeBlind questions, and in the embedding space of Qwen2.5-VL-7B, where the two datasets are highly separable (AUC 0.97; Figure 2b,c).

As a non-contrastive reference, we also evaluate on Video-MME [7], which contains 2,700 four-way multiple-choice questions over 900 internet videos and is scored using standard item accuracy. Video-MME measures general video question answering ability, whereas MotionBlind and TimeBlind specifically isolate temporal reasoning through contrastive evaluation.

Evaluation Metric. MotionBlind adopts the same evaluation protocol as TimeBlind [13], which extends the contrastive evaluation framework introduced by Winoground [25]. Each instance forms a 2×2 contrastive task consisting of two near-identical videos (i_0, i_1) and two complementary questions (q_0, q_1). The ground-truth answers follow a diagonal pattern: q_0 is true for i_0 and false for i_1 , while q_1 is false for i_0 and true for i_1 , yielding four binary prediction items per instance.

Because the paired videos share identical appearance, objects, and scene context while differing only in the motion attribute under evaluation, a model that ignores temporal information cannot satisfy the complete 2×2 pattern, regardless of how strong its object recognition or language priors are.

We report four nested evaluation metrics of increasing strictness. Acc measures the fraction of correctly answered items independently (chance 50%) and is the most permissive metric. Q_{Acc} and V_{Acc} require both items associated with a question or a video, respectively, to be answered correctly (chance 25%). Our primary metric is I_{Acc} , which counts an instance as correct only if all four prediction items are correct simultaneously (chance 6.25%). This strict instance-level metric measures complete understanding of the contrastive motion relationship and prevents models from receiving partial credit by exploiting appearance or language biases.

Human Evaluation. To establish the human ceiling and verify independent answerability, we conducted an internal study with five annotators. Each annotator evaluated all 60 instances. Crucially, clips were presented *one at a time* (not side-by-side) to ensure that the ground-truth answer for each clip could be determined from its motion alone, without requiring direct visual comparison to its paired clip. Annotators achieved a mean I_{Acc} of 91.3% (range 85–97%) and a per-item Acc of 97.8%, confirming that the benchmark is trivially solvable by humans when the video is present and ordered.

4 Experimental Setup

Models. We evaluate six open Video-LLMs chosen to span training emphases: Eagle2.5-8B, a long-context model optimized for processing high-frame-count video sequences; Qwen3-VL-4B, a recent general model that encodes time as textual timestamp tokens prefixed to each temporal patch; Motion-o (7B), a motion-tuned variant of Qwen2.5-VL-7B, whose backbone instead ties temporal position IDs to absolute time; Gemma-4-12B-it, a larger general model with a native video path; Molmo2-8B, an open video-native model; and Inkling-Small, a reasoning model (evaluated with reasoning disabled to match the benchmark’s no-CoT convention). We additionally evaluate two frontier proprietary models (Gemini 3.1 Pro, GPT-5.6 Luna) as reference points. All models are treated as fixed perceptual front ends; we do not fine-tune.

Controlled axes. We vary three levers while holding the model fixed. (i) *Frame budget*: $N \in \{1, 4, 8, 16, 24\}$ frames. (ii) *Frame selection*: uniform (evenly spaced), random, HORNet (a GRPO-trained learned selector), and Frame2Clip (training-free key-clip selection); for HORNet we use the released short-video checkpoint. (iii) *Integrity probes* at a matched 16-frame budget: *no-video* text-only, *shuffled*-frames (content preserved, order destroyed), and *reversed*-frames.

Implementation Details. We append “Please output Yes or No.” to every yes/no question (for TimeBlind’s multiple-choice items, “Please output A or B.”) and parse an explicit “Answer: yes/no” (resp. A/B) or, failing that, the first standalone yes/no (A/B); unparseable outputs are scored incorrect. For Video-MME we instead use its standard four-way prompt (“Respond with only the letter. . .”) and extract the chosen option with ordered patterns, under the same unparseable-scores-wrong rule. All models use greedy decoding ($T=0.0$). Molmo2-8B and Qwen3-VL-4B run the default zero-shot prompt at 128 tokens; Eagle2.5-8B and Motion-o, which degrade under constrained generation, use their authors’ recommended chain-of-thought scaffolds (300 tokens, repetition penalty 1.2).

5 Experimental Results

Across every model, frame budget, and sampling strategy we test, motion understanding is not solved, and the levers that help most are the ones that confirm, rather than close, the gap.

Open models fail; only a frontier model breaks away. Table 1 reports the headline numbers. Among open models the picture is uniform: the best, Eagle2.5-8B, reaches only 11.7% I_{Acc} on MotionBlind (Qwen3-VL-4B, strongest on TimeBlind at 24.7%, manages 8.3%); the rest span 10.0% down to 3.3%, with a larger 12B parameter model (Gemma-4-12B-it) landing among the weakest at 3.6% on the 56-instance subset, so scale alone does not help. Only the frontier Gemini 3.1 Pro breaks away, at 60.0% I_{Acc} (80.8% per-item Acc); GPT-5.6 Luna stays at 15.0%, near the open-model band. Though Gemini performs better relative to other tested models, it still remains well short of the human ceiling (91.3% I_{Acc} , 97.8% Acc; five annotators) and, as we show below, collapses on the hardest physical categories. For the open models Acc sits in the 52–59% band even when I_{Acc} is at the floor, certifying fluent guessing on a balanced yes/no set rather than the ability to tell the two clips apart. This gap between Acc and I_{Acc} is the illusion the benchmark is named for (the open models we test on it reach 58–70% on Video-MME in their best configuration, Table 1; full budget in Figure 8 and Table 5, Appendix C).

The task genuinely requires video and temporal order. A key question is whether models succeed by understanding motion or by exploiting appearance and language shortcuts. To answer this, we evaluate three integrity probes across four models and three datasets: (i) shuffled frames, (ii) reversed frames, and (iii) a text-only setting in which the video is removed entirely. The first two probes preserve the visual content of individual frames while disrupting temporal relationships, whereas the third tests whether the prompted questions can be answered from language alone. Table 2 shows that MotionBlind is highly dependent on both video content and correct temporal ordering. Removing the video or disrupting frame order reduces every model’s I_{Acc} to at or near 0%, indicating that the benchmark cannot be solved reliably without reasoning over motion across frames. The same probes produce a very different outcome on the comparison benchmarks. On Video-MME, shuffling frames reduces Acc by only 0.4 to 1.9 points, reversing frames by 1.0 to 2.5 points, and removing the video entirely still leaves models with 39 to 44% accuracy, substantially above the 25% chance level. This suggests that many Video-MME questions can be answered from appearance and language priors alone. TimeBlind behaves differently: removing the video reduces performance to near 0% I_{Acc} , confirming that the benchmark cannot be solved from language alone, while shuffled and reversed frames still yield approximately chance-level performance (6.25%), indicating that some instances remain solvable without preserving the correct temporal order.

Together, these results show that MotionBlind simultaneously eliminates language leakage and enforces temporal reasoning. Success requires both the video and an understanding of the temporal relationships between frames, making motion perception a prerequisite for solving the benchmark.

The gap is concentrated in physical magnitude. Breaking I_{Acc} down by category (Figure 4) localizes the failure. For every open model, *speed* and *magnitude* sit at zero I_{Acc} ; only *direction* (translational) carries any signal (22–33%). Gemini 3.1 Pro is the exception, clearing chance on three

<i>Best configuration per model across all frame sweeps</i>					
Model	MotionBlind		TimeBlind		Video-MME
	I_{Acc}	Acc	I_{Acc}	Acc	Acc
<i>Chance</i>	6.25	50.0	6.25	50.0	25.0
<i>Open Video-LLMs</i>					
Eagle2.5-8B [4]	11.7	58.8	24.2	66.5	65.4
Qwen3-VL-4B [19]	8.3	58.3	24.7	65.3	59.7
Motion-o (7B) [8] [‡]	6.7	55.4	16.3	61.0	58.2
Gemma-4-12B-it [23] [*]	3.6	52.2	14.3	60.2	70.3
Molmo2-8B [2, 5]	3.3	54.2	12.2	59.0	60.9
Inkling-Small [24] [†]	10.0	57.1	—	—	—
<i>Frontier proprietary</i>					
Gemini 3.1 Pro [9]	60.0	80.8	48.2	76.2	78.2
GPT-5.6 Luna [18] [¶]	15.0	62.1	46.3	77.3	74.8
<i>Human</i> [§]	91.3	97.8	98.2	99.3	87.9

Table 1: Open models fail; only a frontier model breaks away. Instance Accuracy (I_{Acc} ; chance 6.25%) and per-item Acc across the three suites; Video-MME is 4-way multiple choice with no paired structure, so it reports Acc only (chance 25%). On MotionBlind, Acc hovers near the 50% coin-flip even when I_{Acc} is at the floor: models look competent item-by-item but cannot tell the paired clips apart, while the same open models clear Video-MME comfortably (58–70% against 25% chance). Scores are each model’s best combination of frame budget and sampling method per dataset. ^{*}Evaluated on a representative 56-instance subset (224 questions) of the 60-instance suite; not 1:1 comparable. [†]reasoning model, evaluated with reasoning disabled on the 60-instance set (provisional at $n=60$). [‡]motion-tuned Qwen2.5-VL-7B. [¶]GPT-5.6 Luna at 720×1280, detail=high. [§]MotionBlind human is the mean of 5 annotators in our study (60 instances each; range 85–97 I_{Acc}); TimeBlind human from [13]. ^{||}The Gemini 3.1 Pro TimeBlind result is reported from Gemini 3.0 in the original paper; Gemini 3.0 is now deprecated, so we were unable to rerun the evaluation on the same model. “—” = not computed for this work.

<i>Matched budget: uniform, N=16</i>				
Model	Ord.	Shuf.	Rev.	Vid.
<i>MotionBlind (I_{Acc})</i>				
Eagle2.5-8B	10.0	0.0	0.0	0.0
Qwen3-VL-4B	6.7	0.0	0.0	0.0
Motion-o	6.7	0.0	0.0	0.0
Molmo2-8B	1.7	0.0	1.7	0.0
<i>TimeBlind (I_{Acc})</i>				
Eagle2.5-8B	21.3	7.3	5.3	0.0
Qwen3-VL-4B	23.7	7.5	4.5	0.0
Motion-o	13.0	4.0	2.0	0.3
Molmo2-8B	10.3	5.7	3.8	0.0
<i>Video-MME (Acc; chance 25)</i>				
Eagle2.5-8B	62.2	61.3	59.9	43.7
Qwen3-VL-4B	56.6	54.9	54.6	41.6
Motion-o	54.9	54.5	53.9	39.4
Molmo2-8B	59.8	57.9	57.3	44.3

Table 2: Integrity probes at a matched uniform-16 frame budget. We evaluate each benchmark on the original video, shuffled/reversed frames, and text-only input. On *MotionBlind*, removing video or disrupting temporal order reduces all models to at or near 0% I_{Acc} , below the 6.25% chance level, showing that success requires both visual evidence and correct temporal ordering. On *TimeBlind*, disrupting order causes a 2–4× drop, while models retain 2–7.5% I_{Acc} , indicating that some questions remain answerable without full temporal reasoning. In contrast, on *Video-MME*, the same temporal order probes reduce performance by only 0.4–2.5 accuracy points, with text-only performance remaining at approximately 39–44%, well above the 25% chance level. Together, these results show that MotionBlind is uniquely dependent on temporal motion understanding with negligible language or prompt leakage.

of four categories, yet it too drops to 14% on *speed*, the one quantity defined purely by rate. The models are not uniformly blind to time; they specifically fail to encode how fast and how far, exactly the kinematics a world model must predict.

More frames help, then plateau far below solving. Since the models are order-sensitive, does simply showing more of the clip help? Figure 3 sweeps the frame budget under each selection strategy. Accuracy rises with N and saturates around $N=16$ – 24 for every model, but the ceiling is low: Eagle plateaus near 12% and the others at or below $\sim 8\%$. Frames are a lever with a hard stop: they let the model see more of a motion it still cannot represent.

Dynamic frame selection does not rescue it. If uniform sampling saturates, perhaps a smarter selector would spend the budget better. It does not. On the flagship Eagle model, the GRPO-trained learned selector (HORNet) *underperforms* plain uniform sampling at a matched 16-frame budget (10.0 uniform vs. 8.3 HORNet on MotionBlind), while random sampling merely matches it; HORNet occasionally nudges a weaker model’s number up or down, but no strategy pushes Eagle above 11.7% at any budget (with Molmo2 never above 3.3%). The learned selector offers no lever here for a diagnosable reason: on this set its frame keep-probabilities are nearly flat (near-zero variance on 76 of 82 clips), so its top- K picks are close to arbitrary rather than a tuned, motion-aware selection; Figure 7 shows which frames each selector keeps and full-budget in Figure 5. Following Sun et al. [22], we pool F2C candidates at 1 FPS. MotionBlind clips average 5.9 s, so the pool is smaller than K_{anchor} on 73% of clips at $N=8$ and on all 82 at $N=24$: F2C returns the full pool rather than a selection, and the

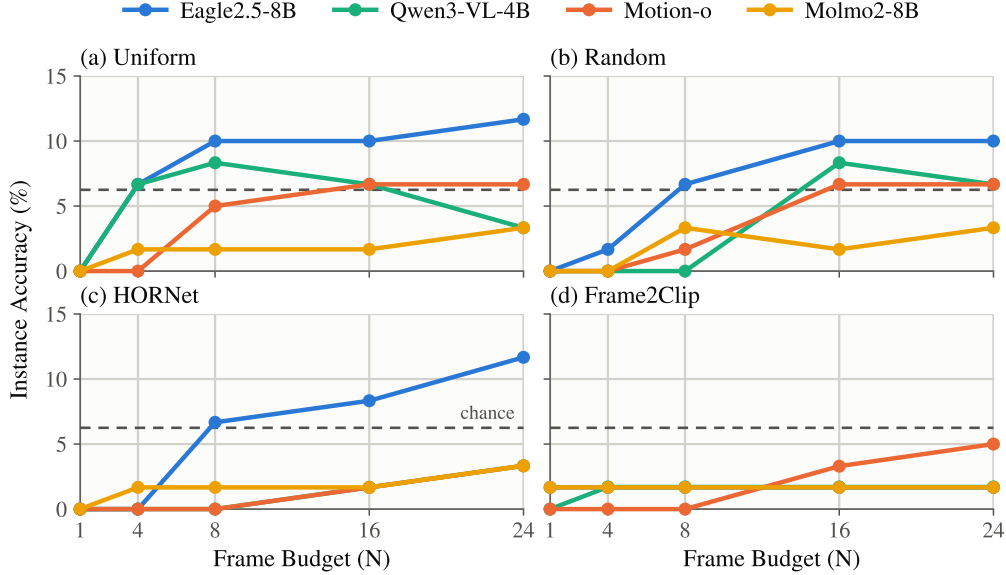


Figure 3: More frames help, then plateau far below solving. MotionBlind I_{Acc} vs. frame budget N under each frame-selection strategy: (a) uniform, (b) random, (c) HORNet, (d) Frame2Clip. The dashed line marks chance (6.25%). Every open model saturates by $N=16-24$ and plateaus at or below 12% for every selector. In (d), F2C is flat by construction: F2C’s 1 FPS pool saturates on second-scale clips (Sec. 5).

anchor sets at $N=16$ and $N=24$ are identical. The F2C column is therefore a scope boundary rather than evidence about frame selection, and we report it as such. Changing which frames are shown does not change whether motion is read (all samplers $\times$ budgets, with exact numbers, in Table 3).

6 Discussion: Implications for World Models

The through-line is that no post-hoc lever (more frames, smarter selection, better prompting) recovers a signal the representation never encoded, and even the one frontier model that clears MotionBlind overall still fails on physical magnitude. This matters for the world model stack: a Video-LLM used to auto-label transitions, score success for model-based RL, or ground instructions is trusted to read exactly the motion it here fails to read. A reward model that cannot tell a gentle placement from a forceful slam supplies noisy or inverted supervision on the physical variables that matter most, and because Acc looks healthy the failure is easy to miss. The ~ 19 -point text-only floor on Video-MME is the quantified form of this illusion: a benchmark can look healthy while the model reads no motion at all. Two implications follow: pipelines that use learned keyframe selection to cut cost inherit the selector’s blind spots, removing the temporal coverage a physical predictor needs; and MotionBlind’s contrastive, order-sensitive structure is itself a training signal; a pairwise, order-aware objective that rewards telling i_0 from i_1 targets the gap that appearance-driven pretraining leaves open.

6.1 Limitations

Our study has limitations. MotionBlind’s 60 self-recorded instances trade scale for controlled isolation of physical variables, so results carry small-sample variance; all clips feature a single actor indoors, leaving in-the-wild and egocentric generalization open; and constrained yes/no parsing, while removing free-form ambiguity, may understate reasoning a model cannot verbalize in binary form. Our open-weight models span $\leq 12B$ parameters, so we cannot fully separate architectural from scale limits, though a larger model not closing the gap is suggestive.

6.2 Broader Impact and Ethics

MotionBlind is a diagnostic for failure modes that matter in safety-critical physical AI and robotics, where misreading force or speed could cause unsafe actions. All video was collected with the participant’s informed consent, faces are automatically blurred prior to release, and the dataset contains no sensitive or personally identifiable information.

7 Conclusion

We introduced MotionBlind, a self-recorded contrastive benchmark isolating physically grounded motion perception, and used it with TimeBlind to study six open Video-LLMs and two frontier models. The conclusion is consistent: for open models motion understanding is not solved, the residual signal is genuinely temporal, and neither more frames, larger models, nor dynamic selection closes the gap; even the frontier model that clears the benchmark overall fails on physical magnitude. A front end that cannot tell two speeds of the same action apart is not yet a trustworthy foundation for a world model. Its taxonomy also specifies further conditions for future collection: mass, acceleration, and a world-frame setting where the camera tracks the object; probing whether the perception inference gap widens as the inferred quantity moves further from the visible motion.

References

- [1] Niket Agarwal, Arslan Ali, Maciej Bala, Yogesh Balaji, Erik Barker, Tiffany Cai, Prithvijit Chattopadhyay, Yongxin Chen, Yin Cui, Yifan Ding, et al. Cosmos world foundation model platform for physical ai, 2025. URL <https://arxiv.org/abs/2501.03575>.
- [2] Allen Institute for AI (AI2). Molmo 2: State-of-the-art video understanding, pointing, and tracking. <https://allenai.org/blog/molmo2>, 2025.
- [3] Xiangyu Bai, Bishoy Galoaa, and Sarah Ostadabbas. Horner: Task-guided frame selection for video question answering with vision-language models. *arXiv preprint arXiv:2603.18850*, 2026.
- [4] Guo Chen, Zhiqi Liu, et al. Eagle 2.5: Boosting long-context post-training for frontier vision-language models. *arXiv preprint arXiv:2504.15271*, 2025.
- [5] Matt Deitke, Christopher Clark, Sangho Lee, Rohun Tripathi, Yue Yang, Jae Sung Park, Mohammadreza Salehi, Niklas Muennighoff, Kyle Lo, Luca Soldaini, et al. Molmo and pixmo: Open weights and open data for state-of-the-art multimodal models. *arXiv preprint arXiv:2409.17146*, 2024.
- [6] Haoan Feng, Sri Harsha Musunuri, and Guan-Ming Su. Geometry-guided camera motion understanding in videoLLMs. *arXiv preprint arXiv:2603.13119*, 2026.
- [7] Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal LLMs in video analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [8] Bishoy Galoaa, Shayda Moezzi, Xiangyu Bai, and Sarah Ostadabbas. Motion-o: Trajectory-Grounded Video Reasoning. *arXiv preprint arXiv:2603.18856*, 2026.
- [9] Google DeepMind. Gemini 3.1 Pro model card. <https://storage.googleapis.com/deepmind-media/Model-Cards/Gemini-3-1-Pro-Model-Card.pdf>, February 2026.
- [10] David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018.
- [11] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations (ICLR)*, 2020.
- [12] Jongseo Lee, Hyuntak Lee, Sunghun Kim, Sooa Kim, Jihoon Chung, and Jinwoo Choi. Which way did it move? diagnosing and overcoming directional motion blindness in video-LLMs. *arXiv preprint arXiv:2605.22823*, 2026.
- [13] Baiqi Li, Kangyi Zhao, Ce Zhang, Chancharik Mitra, Jean de Dieu Nyandwi, and Gedas Bertasius. Timeblind: A spatio-temporal compositionality benchmark for video llms. *arXiv preprint arXiv:2602.00288*, 2026.
- [14] Yuanxin Liu, Shicheng Li, Yi Liu, Yuxiang Wang, Shuhuai Ren, Lei Li, Sishuo Chen, Xu Sun, and Lu Hou. Tempcompass: Do video LLMs really understand videos? In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

- [15] Shayda Moezzi, Umer Saleem, Andong Deng, Chen Chen, and Sarah Ostadabbas. Position: Video llms must not ignore the pixel dynamics in plain sight. In *Forty-third International Conference on Machine Learning Position Paper Track*, 2026.
- [16] Burt L. Monroe, Michael P. Colaresi, and Kevin M. Quinn. Fightin’ words: Lexical feature selection and evaluation for identifying the content of political conflict. *Political Analysis*, 16(4):372–403, 2008.
- [17] NVIDIA et al. Cosmos 3: Omnimodal world models for physical ai, 2026. URL <https://arxiv.org/abs/2606.02800>.
- [18] OpenAI. GPT-5.6 system card. <https://deploymentsafety.openai.com/gpt-5-6>, July 2026.
- [19] Qwen Team, Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, et al. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*, 2025.
- [20] Qwen Team, Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- [21] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- [22] Guangyu Sun et al. From frames to clips: Training-free adaptive key clip selection for long-form video understanding. *arXiv preprint arXiv:2510.02262*, 2025.
- [23] Gemma Team. Gemma 4 technical report, 2026. URL <https://arxiv.org/abs/2607.02770>.
- [24] Thinking Machines Lab. Inkling model card. <https://thinkingmachines.ai/model-card/inkling/>, July 2026.
- [25] Tristan Thrush, Ryan Jiang, Max Bartolo, Amanpreet Singh, Adina Williams, Douwe Kiela, and Candace Ross. Winoground: Probing vision and language models for visio-linguistic compositionality. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [26] Jianrui Zhang, Mu Cai, and Yong Jae Lee. Vinoground: Scrutinizing llms over dense temporal reasoning with short videos. *arXiv preprint arXiv:2410.02763*, 2024.

A Per-category breakdown

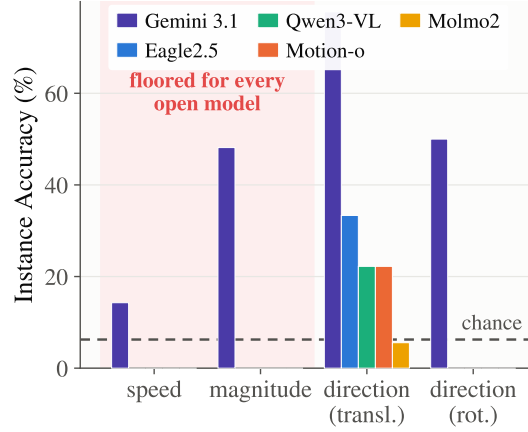


Figure 4: Per-category I_{Acc} on MotionBlind (matched uniform-16 budget). For every open model, *speed* and *magnitude* are at the chance floor and only *direction (translational)* carries signal; Gemini 3.1 Pro clears three of four categories yet still fails on *speed*.

B Full frame-selection sweeps

Per-selector budget sweeps for every open model on MotionBlind (Figure 5) and TimeBlind (Figure 6); all use the shared sampler key.

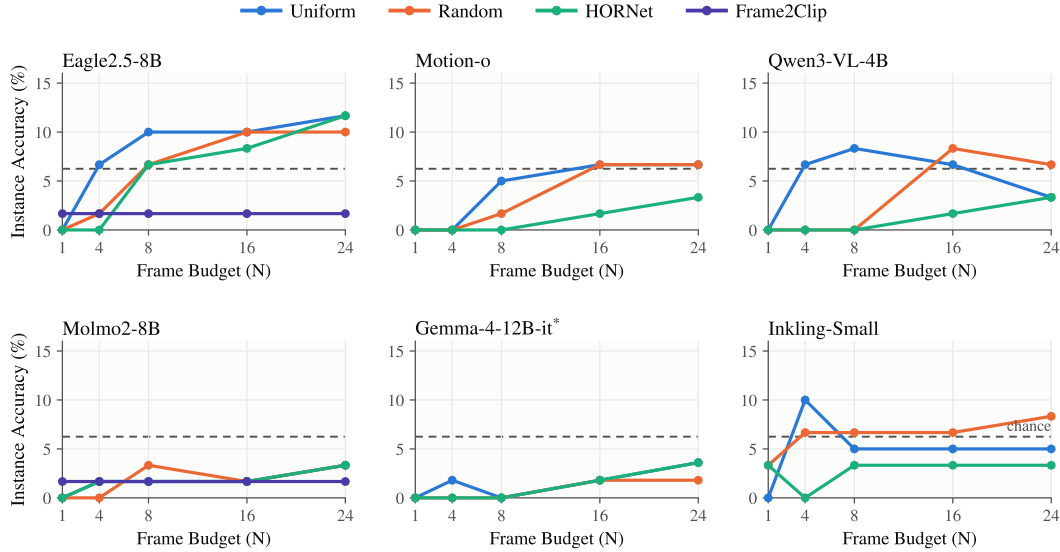


Figure 5: MotionBlind I_{Acc} vs. frame budget N , one line per frame selector. Every open model plateaus in the low teens or below and no selector escapes the floor. *Gemma-4-12B-it is on the MotionBlind-82 cache subset (not 1:1 comparable).

C Video-MME sanity check

Table 2 (Video-MME panel) shows why the non-contrastive reference matters: shuffling frames moves accuracy by only 0.4–1.9 points, reversing by 1.0–2.5, and removing the video entirely still leaves 39–44% against the 25% chance level. Standard multiple-choice video QA is therefore substantially answerable from appearance and language priors alone, whereas MotionBlind’s contrastive protocol removes that leakage (text-only $I_{Acc} = 0$ for every model).

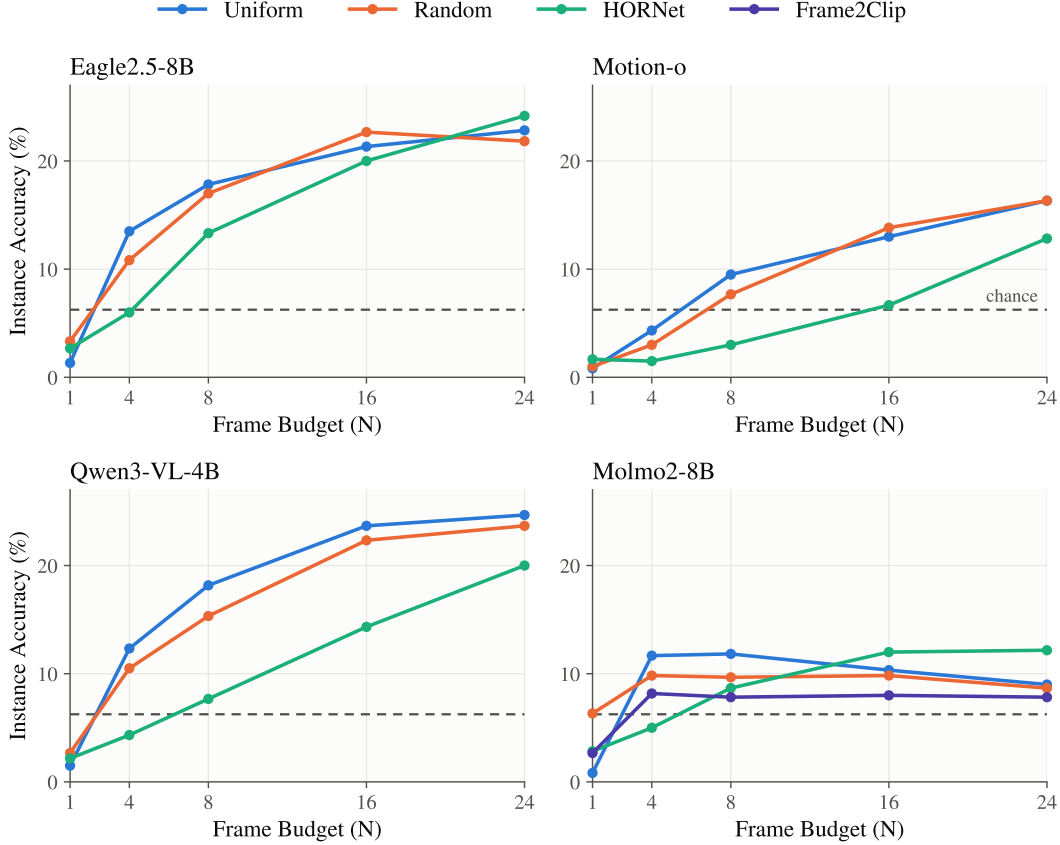


Figure 6: TimeBlind I_{Acc} vs. frame budget N (same models and selectors). Accuracy rises with N but saturates well short of the human ceiling; Molmo2-8B adds a Frame2Clip curve.



Figure 7: Dynamic frame selection on one MotionBlind clip at a budget of $K = 8$. The 32 candidate frames are spaced evenly over the clip; the question is "Does the person stir slower the first time compared to the second time?" (ground truth: no). A colored border marks a frame that a selector returns. HORNet (teal) keeps the 8 frames with the highest keep probability. Frame2Clip (violet) places anchors at the 8 peaks of the SigLIP2 frame–query similarity (white dot) and grows each anchor into a contiguous clip rendered at decreased resolution (checkerboard), so 29 frames cost 7.25 of the 8 native-frame units.

D Full sweep results (all numbers)

The exact per-configuration values behind Figures 5–8 and the frame-selection comparison: every model $\times$ frame selector $\times$ budget N , on MotionBlind (Table 3), TimeBlind (Table 4), and Video-MME (Table 5).

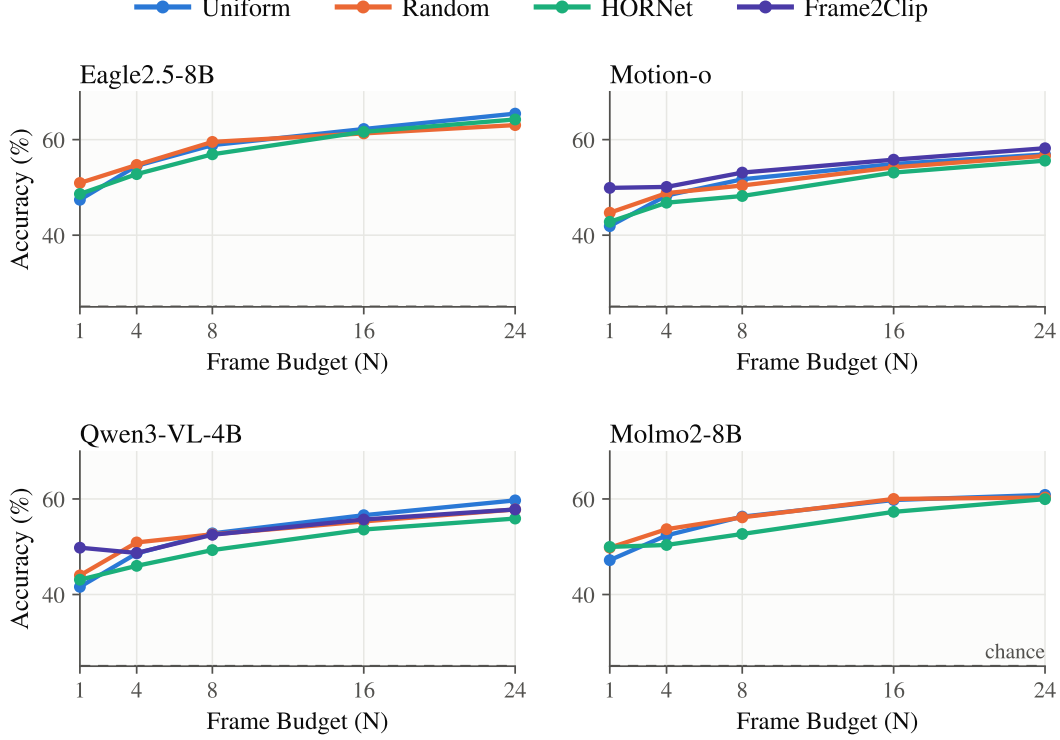


Figure 8: Video-MME accuracy vs. frame budget N (chance 25%). The same models reach ~50–65%, so the MotionBlind failure is motion-specific rather than general incompetence. Eagle2.5-8B and Molmo2-8B are from our runs; Motion-o and Qwen3-VL-4B also include a Frame2Clip curve.

Table 3: Full MotionBlind sweep: Instance Accuracy I_{Acc} (%) for every model, frame selector, and budget N (chance 6.25%). “—” = not run. *Gemma-4-12B-it on the MotionBlind-82 cache subset.

Sampler	$N=1$	4	8	16	24
<i>Eagle2.5-8B</i>					
Uniform	0.0	6.7	10.0	10.0	11.7
Random	0.0	1.7	6.7	10.0	10.0
HORNet	0.0	0.0	6.7	8.3	11.7
Frame2Clip	1.7	1.7	1.7	1.7	1.7
<i>Qwen3-VL-4B</i>					
Uniform	0.0	6.7	8.3	6.7	3.3
Random	0.0	0.0	0.0	8.3	6.7
HORNet	0.0	0.0	0.0	1.7	3.3
Frame2Clip	0.0	1.7	1.7	1.7	1.7
<i>Motion-o (7B)</i>					
Uniform	0.0	0.0	5.0	6.7	6.7
Random	0.0	0.0	1.7	6.7	6.7
HORNet	0.0	0.0	0.0	1.7	3.3
Frame2Clip	0.0	0.0	0.0	3.3	5.0
<i>Gemma-4-12B-it*</i>					
Uniform	0.0	1.8	0.0	1.8	3.6
Random	0.0	0.0	0.0	1.8	1.8
HORNet	0.0	0.0	0.0	1.8	3.6
<i>Molmo2-8B</i>					
Uniform	0.0	1.7	1.7	1.7	3.3
Random	0.0	0.0	3.3	1.7	3.3
HORNet	0.0	1.7	1.7	1.7	3.3
Frame2Clip	1.7	1.7	1.7	1.7	1.7
<i>Inkling-Small</i>					
Uniform	0.0	10.0	5.0	5.0	5.0
Random	3.3	6.7	6.7	6.7	8.3
HORNet	3.3	0.0	3.3	3.3	3.3

Table 4: Full TimeBlind sweep: Instance Accuracy I_{Acc} (%) (chance 6.25%).

Sampler	$N=1$	4	8	16	24
<i>Eagle2.5-8B</i>					
Uniform	1.3	13.5	17.8	21.3	22.8
Random	3.3	10.8	17.0	22.7	21.8
HORNet	2.7	6.0	13.3	20.0	24.2
<i>Qwen3-VL-4B</i>					
Uniform	1.5	12.3	18.2	23.7	24.7
Random	2.7	10.5	15.3	22.3	23.7
HORNet	2.2	4.3	7.7	14.3	20.0
<i>Motion-o (7B)</i>					
Uniform	0.8	4.3	9.5	13.0	16.3
Random	1.0	3.0	7.7	13.8	16.3
HORNet	1.7	1.5	3.0	6.7	12.8
<i>Gemma-4-12B-it</i>					
Uniform	0.3	9.8	12.3	13.3	14.3
Random	1.3	6.7	11.7	14.0	13.3
HORNet	1.5	3.7	7.0	11.0	13.5
<i>Molmo2-8B</i>					
Uniform	0.8	11.7	11.8	10.3	9.0
Random	6.3	9.8	9.7	9.8	8.7
HORNet	2.8	5.0	8.7	12.0	12.2
Frame2Clip	2.7	8.2	7.8	8.0	7.8

Table 5: Full Video-MME sweep: accuracy (%) on the standard benchmark (chance 25%; majority baseline 27.2%). [†]Motion-o, Qwen3-VL-4B, and Gemma-4-12B-it Video-MME are the authors’ cited numbers.

Sampler	$N=1$	4	8	16	24
<i>Eagle2.5-8B</i>					
Uniform	47.4	54.5	58.9	62.2	65.4
Random	50.9	54.7	59.5	61.3	63.0
HORNet	48.6	52.8	56.9	61.6	64.2
<i>Qwen3-VL-4B[†]</i>					
Uniform	41.6	48.6	52.8	56.6	59.7
Random	44.0	50.9	52.6	55.3	57.8
HORNet	43.1	46.0	49.3	53.6	55.9
Frame2Clip	49.8	48.7	52.5	55.7	57.8
<i>Motion-o (7B)[†]</i>					
Uniform	41.9	48.3	51.7	54.9	56.9
Random	44.7	48.8	50.4	54.2	56.6
HORNet	42.8	46.8	48.2	53.1	55.6
Frame2Clip	49.9	50.1	53.1	55.8	58.2
<i>Gemma-4-12B-it[†]</i>					
Uniform	35.9	45.4	48.6	52.1	54.1
Random	38.9	43.5	47.6	50.7	52.1
HORNet	35.3	42.5	46.2	51.0	53.7
<i>Molmo2-8B</i>					
Uniform	47.2	52.4	56.3	59.8	60.9
Random	49.8	53.7	56.1	60.0	60.3
HORNet	50.0	50.4	52.7	57.3	60.0